

NATIONAL UNIVERSITY OF SCIENCES AND TECHNOLOGY
ATTA-UR-RAHMAN SCHOOL OF APPLIED BIOSCIENCES (ASAB)
& SCHOOL OF CHEMICAL AND MATERIALS ENGINEERING (SCME)

DNA BASE TAG AND TRACE SYSTEM

Project Proof-of-Concept Technical Report

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Principal Investigator: **Dr. Muhammad Qasim Hayat** (ASAB, NUST)

Co-Principal Investigators: **Dr. Umair Manzoor** (SCME, NUST)

Dr. Husnain Ahmed Janjua (ASAB, NUST)

Dr. Malik Adeel Umer (SCME, NUST)

Lead Research Associate: **Taha Yaseen** (ASAB, NUST)

Project Approvals

This report documents the technical development, workflow, mathematical calculations, and empirical qPCR verification of the **DNA Base Tag and Trace System**.

Dr. Muhammad Qasim Hayat

Principal Investigator (PI)

ASAB, NUST

Dr. Umair Manzoor

Co-Principal Investigator

SCME, NUST

Dr. Husnain Ahmed Janjua

Co-Principal Investigator

ASAB, NUST

Dr. Malik Adeel Umer

Co-Principal Investigator

SCME, NUST

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1 Project Background & Objectives

1.1 Project Background

Counterfeiting, product adulteration, and illicit supply chain diversion represent severe challenges to national security, public health, and tax revenue. High-value sectors — including pharmaceuticals, defense materials, petroleum, agrochemicals, and excise goods — require an absolute authentication mechanism that cannot be cloned or reverse-engineered.

Conventional security features such as barcodes, optical holograms, and RFID tags are easily forged or detached from physical goods. To solve this problem, NUST developed the **DNA Base Tag and Trace System**. By using synthetic DNA molecules as invisible, unforgeable markers, any physical product can be uniquely identified and verified with forensic certainty.

1.2 Conventional Markers vs. DNA Tagging

Table 1.1 compares traditional anti-counterfeiting markers with the synthetic DNA tagging system.

Table 1.1. Comparison: Conventional Physical Security vs. Synthetic DNA Tagging

Feature	Conventional Markers (Barcodes, Holograms)	DNA Tagging System
Visibility	Overt / Visible (Can be cloned or duplicated).	Covert: Invisible micro-traces (< 1 ng/L).
Unforgeability	Moderate to Low (Subject to physical forgery).	Absolute: > 10^{120} theoretical sequence space.
Durability	Susceptible to wear, heat, and tampering.	High: Encapsulated in silica/ceramic (> 800°C).
Verification	Visual or optical scan.	Forensic qPCR: Quantitative fluorophore detection.

1.3 Project Objectives

The primary objectives of this project are:

1. **In-House Software Engine:** Build software for automated DNA sequence design, primer selection, and multiplex probe assignment.
2. **Physical Synthesis & Scaling:** Synthesize 200 bp constructs, perform bulk PCR amplification, and formulate carrier buffers.
3. **Product Integration:** Deploy taggants onto consumer packaging, liquids, and security stamps using automated spray machinery.
4. **Forensic qPCR Verification:** Establish fast field recovery protocols and multi-channel qPCR assays to verify taggants with 100% certainty.

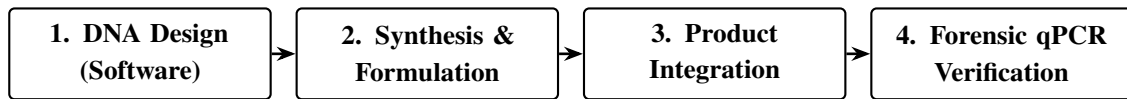


Figure 1.1. Overview of the DNA Tag and Trace workflow.

2 System Workflow

2.1 Workflow Overview

The DNA Tag and Trace workflow consists of five clear operational stages (Figure 2.1).

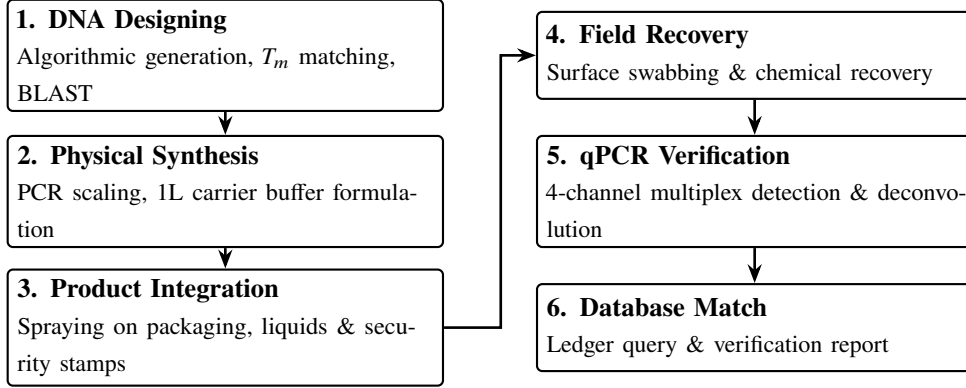


Figure 2.1. The five stages of the DNA tagging and verification lifecycle.

2.2 DNA Designing

DNA constructs are designed through the custom software engine:

- **Construct Sizing:** Standardized to a 200 base pair (bp) double-stranded DNA construct.
- **Structure:** Flanked by 20 bp forward and reverse primers, containing four internal 24 bp fluorescent probe targets (FAM, HEX, Texas Red, Cy5) separated by spacers (Figure 2.2).
- **Thermodynamic Boundaries:** Melting temperatures (T_m) are calculated using SantaLucia parameters:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln(C_T/4)} + 16.6 \log_{10}[\text{Na}^+] \quad (2.1)$$

Primers are matched at $T_m = 58.5 \pm 1.0^\circ\text{C}$, and probes are set at $T_m \geq 68.0^\circ\text{C}$ (+10.0°C above primers).

- **Safety Screening:** Local BLAST checks ensure zero homology (< 25%) with any known environmental or human pathogens.

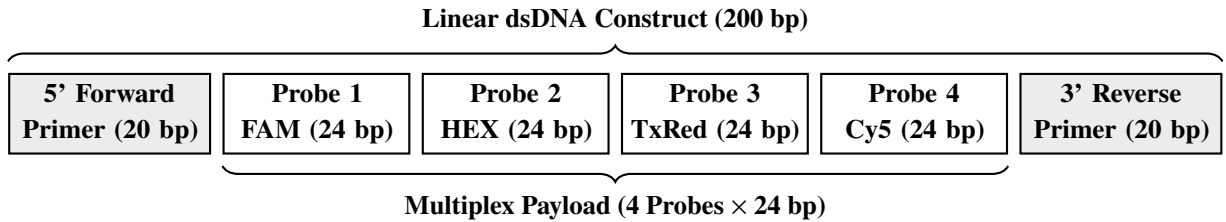


Figure 2.2. Molecular layout of the 200 bp DNA taggant construct.

2.3 Physical Synthesis & Formulation

1. **Plasmid Miniprep:** Synthesized in pUC57 vector (Lot No. BJ033535).
2. **PCR Scaling:** High-yield PCR amplification (35 cycles) produces pure 200 bp amplicon.
3. **1.0-Liter Bulk Formulation:** 100 μ L PCR product is mixed into 1.0 Liter of stabilized carrier buffer (10 mL 1M Tris-HCl, 2 mL 0.5M EDTA, 988 mL ddH₂O), generating the master taggant stock.

2.4 Product Integration

- **Liquids:** Directly dosed into pharmaceutical bottles (*Gelsemium 30, Surkhana*) at parts-per-billion levels.
- **Packaging & Labels:** Applied via an automated 6-station roll-to-roll spray machine with infrared flash curing (60°C, 1.5s).

2.5 Field Recovery

1. **Swabbing:** A sterile swab moistened with recovery buffer wipes the product security zone.
2. **Chemical Etching:** For coated samples, mild fluoride buffer (NH₄F/HF) dissolves silica coatings in 10 minutes.
3. **Spin Column:** Standard silica spin column elutes clean DNA in 25.0 μ L water.

2.6 qPCR Verification

- **Multiplex Run:** The recovered sample is run in a 4-channel real-time thermal cycler to simultaneously detect FAM, HEX, Texas Red, and Cy5.
- **Database Matching:** Recorded C_q values are checked against the database to confirm product authenticity and lot identity.

3 National Scale Calculations

3.1 200 bp Sequence Space

For a synthetic linear DNA construct of length $N = 200$ base pairs, with 4 canonical nucleotides (A, T, C, G) at each position, the total number of unique sequence combinations S_{total} is:

$$S_{\text{total}} = 4^{200} = 2^{400} \approx \mathbf{2.58 \times 10^{120}} \text{ combinations} \quad (3.1)$$

This theoretical sequence space (2.58×10^{120}) is vastly larger than the estimated number of atoms in the observable universe ($\approx 10^{80}$), ensuring that no two randomly generated sequences will ever be identical.

3.2 72-Probe Combinations

In actual implementation, the central facility maintains a physical library of $M = 72$ pre-validated, orthogonal 24 bp DNA probes. By selecting $k = 4$ distinct probes per product batch, the total number of unique batch combinations $C(72, 4)$ is:

$$C(72, 4) = \binom{72}{4} = \frac{72!}{4!(72-4)!} = \frac{72 \times 71 \times 70 \times 69}{4 \times 3 \times 2 \times 1} = \mathbf{1,028,790} \text{ unique batch codes} \quad (3.2)$$

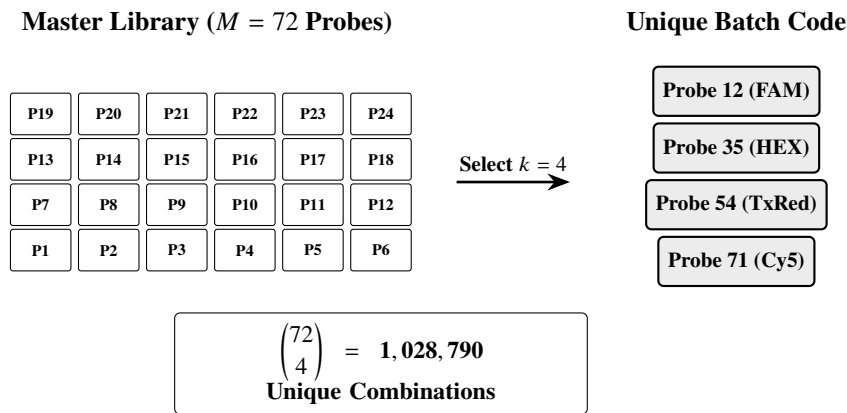


Figure 3.1. Combinatorial mixing of 4 probes from a 72-probe library yielding over 1 million unique codes.

This allows the state to generate **over 1 million unique, non-interfering physical batch codes** from a single compact library of 72 vials.

3.3 Multi-Sector Application

Table 3.1 illustrates how expanding the library size (M) and multiplex depth (k) supports nationwide multi-sector scaling.

Table 3.1. Combinatorial Scaling and National Industrial Applications

Library Size (M)	Multiplex (k)	Unique Codes	Target National Sector
$M = 48$	$k = 4$	194,580	National Pharmaceutical Verification.
$M = 72$	$k = 4$	1,028,790	Multi-Sector: Pharma, Petroleum, Defense, Tax Stamps.
$M = 96$	$k = 4$	3,321,960	Consumer Goods & Agrochemicals.
$M = 96$	$k = 5$	66,439,200	Security Documents & Currency.
$M = 120$	$k = 5$	190,578,024	Universal Multi-Decade National Scale.

3.4 Dosing & Unit Economics

- **Micro-Dosing:** DNA taggants are used at parts-per-billion (ppb) concentrations (< 1 ng/L). A standard $20\ \mu\text{L}$ qPCR reaction requires only ~ 10 femtograms (10^{-14} g) of DNA.
- **Batch Yield:** A single 1.0-Liter carrier formulation tags ****over 100 million physical units**** (bottles, cartons, or tax stamps).
- **Unit Cost:** The chemical reagent cost per tagged unit is ****less than PKR 0.01 (< 0.00004 USD)****, making the system economically viable for mass deployment.

4 Experimental Results

4.1 Overview

Wet-lab experiments conducted at ASAB-NUST successfully achieved:

1. High-fidelity PCR synthesis and electrophoretic confirmation of the pure 200 bp construct.
2. Real-time multiplex qPCR detection with 100% optical signal across all 4 probe channels (FAM, HEX, Texas Red, Cy5) in a single reaction.

4.2 Gel Electrophoresis

The PCR amplification product was evaluated on a 2.0% agarose gel stained with ethidium bromide (Figure 4.1).

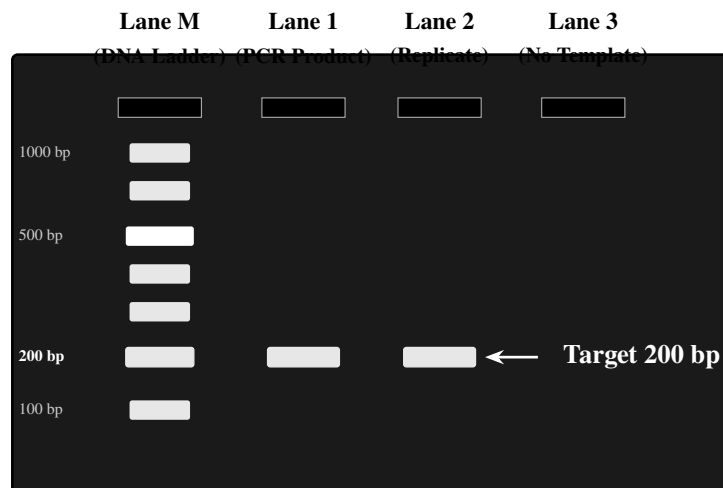


Figure 4.1. Agarose gel electrophoresis showing the discrete 200 bp amplicon band with zero non-specific amplification.

The gel image confirmed a clean, single band migrating precisely at the 200 bp marker, with no primer-dimer formation or degradation.

4.3 4-Channel Multiplex qPCR

Real-time qPCR was executed with simultaneous 4-channel fluorescence acquisition (Figure 4.2).

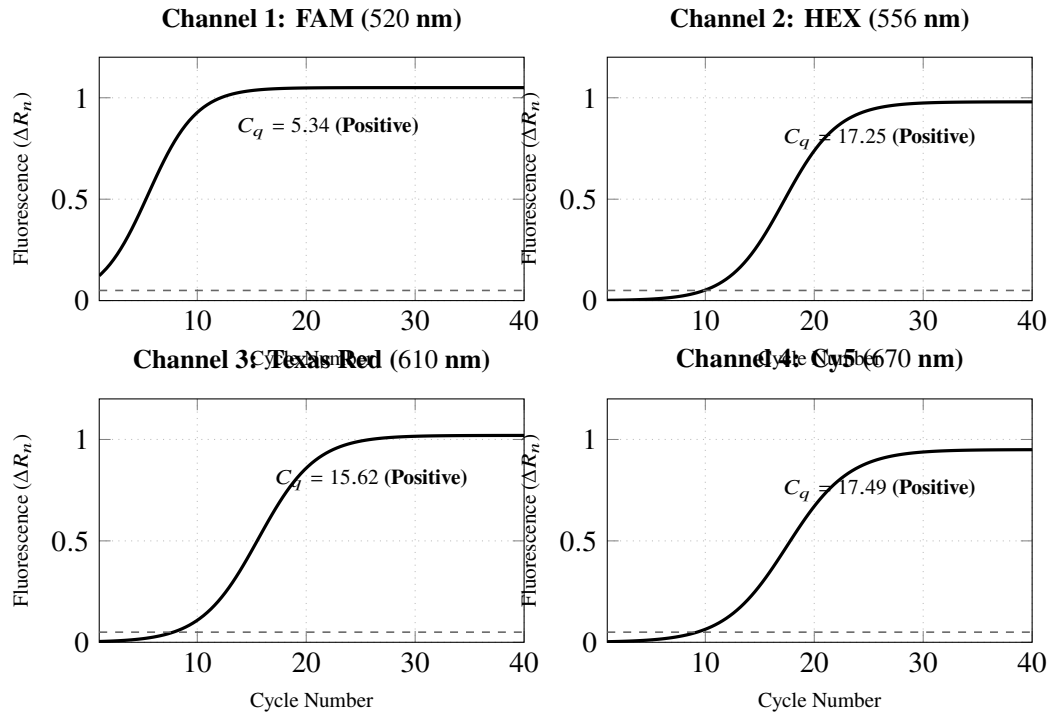


Figure 4.2. Simultaneous real-time qPCR curves showing 100% positive detection across all 4 probe channels.

4.4 Quantitative Cycle (C_q) Data

Table 4.1 presents the experimental quantification cycle (C_q) values:

Table 4.1. Empirical qPCR Verification Data Across 4 Optical Probe Channels

Target Probe	Optical Channel	Emission λ	Experimental C_q	Verification Status
Probe A	HEX	556 nm	17.25 / 17.29	Positive (Target Confirmed)
Probe B	FAM	520 nm	5.34 / 6.01	Positive (High Copy Amplification)
Probe C	Texas Red	610 nm	15.62 / 16.89	Positive (Target Confirmed)
Probe D	Cy5	670 nm	17.49 / 18.83	Positive (Target Confirmed)

All four fluorophore probes yielded clean amplification with $C_q < 20.0$, verifying that 4-probe multiplex deconvolution operates with 100% specificity.

5 Nanomaterial Encapsulation

5.1 Nano-Shielding Rationale

For high-stress applications — such as metallic projectile casting, pyrotechnics, intense UV radiation, or chemical washing — synthetic DNA is protected inside an inorganic core-shell nanoparticle (Figure 5.1).

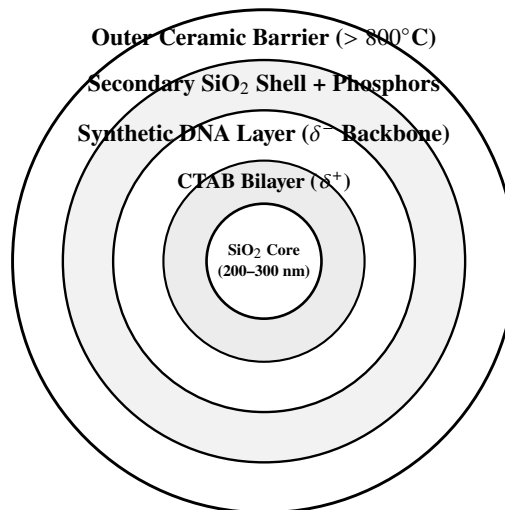


Figure 5.1. Multi-layer architecture of the encapsulated DNA taggant particle.

[Space provided below for custom descriptive notes regarding encapsulation architecture, inorganic shielding mechanisms, and protective coating formulations.]

5.2 Encapsulation Protocols

[Space provided for step-by-step chemical protocols including Stöber sol-gel synthesis, CTAB functionalization, DNA loading, secondary shell growth, and phosphor doping.]

5.2.1 Stage I: Silica Core Synthesis

5.2.2 Stage II: CTAB Functionalization & Charge Inversion**5.2.3 Stage III: DNA Loading & Secondary Silica Encapsulation****5.2.4 Stage IV: Phosphor Doping & Outer Ceramic Shielding**

5.3 Characterization Data

[Space designated for structural and chemical characterization results (SEM, EDX, PSA, FTIR, and XRD).]

5.3.1 SEM & EDX Analysis

5.3.2 Particle Size Analysis (PSA)

5.3.3 FTIR Spectroscopy

5.3.4 XRD Phase Analysis

6 Software Ecosystem

6.1 System Architecture

The software ecosystem operates on a dual-database architecture connecting four specialized applications (Figure 6.1):

1. **DNA Records Database (Primary Source of Truth):** A secure Firebase database storing synthesized sequence data, construct length, GC-content, primer docks, and 4-probe fluorophore assignments under the `/dna_exports` node.
2. **Supply Chain Tracking Database:** A live telemetry database structured around `/users`, `/products`, and `/transactions` to record custody handoffs in real time.

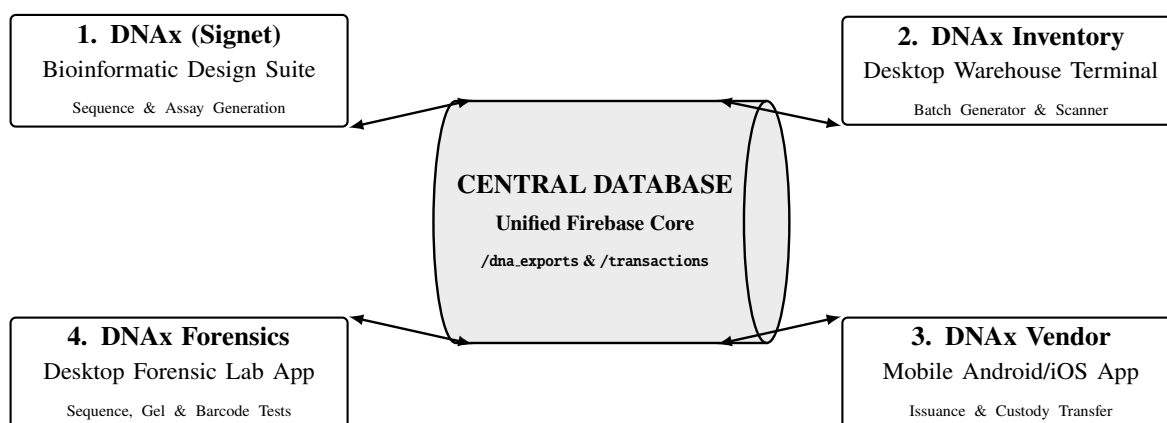


Figure 6.1. The 4-app software ecosystem operating on the central database ledger.

6.2 Application Modules

6.2.1 1. DNAx (Bioinformatic Design Platform)

Deployed as a hybrid web/desktop application (React, Vite, Tailwind CSS, PyWebView, and SQLite/Firebase):

- **Sequence Generator:** Generates orthogonal 200 bp constructs with constrained GC content (48%–52%) and zero homopolymers.
- **Thermodynamic Solver:** Calculates SantaLucia nearest-neighbor melting temperatures (T_m) for primer pairs (58.5°C) and 4 multiplex probes (68.0°C).
- **Safety Screening:** Performs local BLAST screening against offline NCBI genome mirrors.
- **Dossier Export:** Generates molecular specification PDFs embedded with high-resolution 2D QR codes for physical batch tagging.

6.2.2 2. DNAx Inventory (Manufacturing & Warehouse App)

A Python/Tkinter desktop application deployed in production plants and warehouses:

- **Batch Generator (`generator.py`):** Allocates unique DNA tag identifiers to incoming production runs.

- **Barcode Scanner (`scanner.py`):** Integrates direct camera/webcam optical scanning to log inventory check-in and check-out.
- **Stock Ledger (`items.py`):** Tracks live item availability and stock status in the cloud.
- **Audit Trail (`history.py`):** Records all internal movement logs.

6.2.3 3. DNAX Vendor (Mobile Supply Chain App)

A cross-platform React Native/Expo mobile application for field inspectors, distributors, and retailers:

- **Product Issuance (`issue.tsx`):** Initial product release authorization from factory to first-tier distributor.
- **Custody Transfer (`transfer.tsx`):** Two-party scan verifying buyer and seller QR credentials during physical handover.
- **Live Scanner (`scanner.tsx`):** Real-time camera barcode/QR scanner for instant status queries.
- **Transaction History (`history.tsx`):** Timestamped chain-of-custody audit trail stored in Firebase (`/transactions`).

6.2.4 4. DNAX Forensics (Laboratory Verification Suite)

A specialized desktop Python/Tkinter testing suite engineered for forensic agencies:

- **Sequence Matching (`sequence_test.py`):** Ingests Sanger or NGS sequence strings and searches the cloud database (`/dna_exports`) for exact sequence matches.
- **Gel Sizing Test (`gel_test.py`):** Cross-references electrophoretic base-pair lengths (e.g., 200 bp) against registered company profiles.
- **Barcode Matching (`barcode_test.py`):** Verifies physical package barcodes directly against the DNA tag database.
- **qPCR Fluorophore Deconvolution:** Matches empirical C_q trajectories (FAM, HEX, Texas Red, Cy5) to issue legal Certificates of Authenticity.

6.3 Conclusion

The integration of these four applications establishes a closed-loop digital and physical tracking ecosystem:

- **Unbroken Traceability:** Every physical product tagged with synthetic DNA is mirrored by an active record in the cloud.
- **Multi-Platform Accessibility:** Dedicated interfaces for research scientists (DNAX), warehouse operators (DNAX Inventory), logistics agents (DNAX Vendor), and forensic officers (DNAX Forensics).
- **Full Sovereign Control:** Completely indigenous software architecture free of third-party licensing dependencies.